

COMPSCI 3IS3: Information Security

Department of Computing and Software
McMaster University

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Lectures:	Monday 12:30 - 13:20, Tuesday 13:30 - 14:20, Thursday 12:30 - 13:20
Location:	JHE 264
Office Hours:	Monday after the class (By appointment)

Course description

The ubiquity of communication devices and the need for protecting data have made information security an integral element of our everyday lives. Information security in the form of encryption algorithms existed long before computers. However, modern information security has developed far beyond simple encryption algorithms. This course is an introduction to information security with an emphasis on cryptography and applications. We will cover three main topics: we will first talk about basics of information security, then we go over some topics in cryptography, and finally we spend some time on an introduction to network security.

- *Prerequisites:* COMPSCI 2AC3 or 2FA3 or SFWRENG 2FA3, and COMPSCI 2C03 or SFWRENG 2C03.
- *Textbooks:* The lectures will be mainly based on the following references:
 - “Applied Information Security” by Basin, Schaller, Schlapfer
 - “Understanding Cryptography” by Christof Paar and Jan Pelzl. This will be our primary reference, since we will focus a bit more on cryptography.
 - “Security in Computing” by Charles P. Fleeger, Shari Lawrence Pfleeger, and Jonathan Margulies
- *Organization:* 3h lecture.
- *Learning Objectives:* At the end of this course, a successful student will
 - have a reasonable understanding of some of the basic ideas in information security
 - understand the basic mathematical and practical aspects of cryptographic algorithms,
 - be able to explain the cryptographic building blocks of network security protocols.
- *Tentative list of topics:*
 - Part 1: Basics of Information Security
 - Security architecture and infrastructure
 - Security goals (confidentiality, integrity, authenticity, etc.)
 - Part 2: Cryptography

- Stream Ciphers
- The Data Encryption Standard (DES)
- The Advanced Encryption Standard (AES)
- Hash Functions
- Introduction to Public-Key Cryptography
- The RSA Cryptosystem
- Digital Signatures

Part 3: Network Security Protocols

- Overview of networking and PKI
- Transport Layer Security (TLS) protocol
- Overview of SSH

Evaluation

The final grade for this course is the sum of grades for the assignments, quizzes and the final project. There will be four assignments and a final project. The first two assignments will cover the topics related to block ciphers and hash functions. Assignments 3 and 4 will cover public-key cryptography. The final project will be a topic in from the entire course including network security.

Evaluation	% of the final grade	Topic
Assignment 1	10%	Block ciphers and hash functions
Assignment 2	10%	
Assignment 3	20%	Public-key cryptography
Assignment 4	25%	
Final project	30%	A topic from the entire course including Network Security
Quizzes	5%	During class

Evaluation details:

- Quizzes will be given without notice. Students should submit the quiz at the end of the same day. Marks for quizzes will be given equally for effort and solution correctness.
- The assignments will be spread out evenly throughout the semester.
- Late submissions will be penalized at a rate of $2^n\%$, where n is the number of days late. One day late is a 2% penalty, two days 4%, three days 8%, and so on.
- We will use **Avenue to Learn** for course management. All course materials can be accessed through Avenue to Learn. Other things such as announcements, assignment descriptions and submissions, etc, will also be done through Avenue to Learn.
- Dates are subject to change, all changes will be announced in Avenue to Learn.

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/), located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>.

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity/Plagiarism Detection

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster’s use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses With An On-Line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation Of Students With Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Requests For Relief For Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).¹

Academic Accommodation For Religious, Indigenous Or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar’s Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright And Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

¹Updated to the Policy on Requests for Relief for Missed Academic Term Work effective May 1, 2021

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.